

NITHIN KUMAR AITHA

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SUMMARY

Robotics engineer and M.S. Robotics candidate (Dec 2026) who builds ROS 2 perception, SLAM, and sensor fusion systems in C++ and Python on Linux. Built a LiDAR SLAM pipeline producing 0.05 m occupancy grid maps over 500 m of trajectory, and a time-synchronized LiDAR, camera, IMU, and GNSS stack recovering 3D object positions within 0.5 m of GNSS ground truth.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB, Bash

Robotics and Middleware: ROS 2 (Humble), ROS 2 nodes, topics, services, and launch files, TF2, RViz2, Navigation2 (Nav2), slam_toolbox, rosbag, URDF, CMake, colcon

Perception and Estimation: SLAM, simultaneous localization and mapping, LiDAR point cloud processing, occupancy grid mapping, multi-sensor fusion (LiDAR, camera, IMU, GNSS), loop closure detection, pose estimation, state estimation, sensor calibration, camera-LiDAR extrinsic calibration, 2D-to-3D projection, 3D object localization, OpenCV, PCL, Eigen

Machine Learning and Computer Vision: YOLOv8, object detection, MobileNetV2, ResNet50, convolutional neural networks, transfer learning, data augmentation, TensorFlow, Keras

Controls and Planning: Path planning, motion planning, forward and inverse kinematics, Jacobians, quintic trajectory generation, state-space modeling, PID control

Tools and Platforms: Linux (Ubuntu), Git, GitHub, MATLAB, Simulink, ANSYS (FEA), SolidWorks, AutoCAD

ROBOTICS PROJECTS

HD Map Generation and LiDAR-Based SLAM | ROS 2, C++, Python, PCL | github.com/nithinkumaraitha/lidar-slam-hd-mapping

- Engineered a ROS 2 LiDAR SLAM pipeline generating occupancy grid maps at 0.05 m resolution over 500 m of combined indoor and outdoor trajectory.
- Validated loop closure consistency and TF2 frame transformations in RViz2 across 15 recorded rosbag runs, diagnosing and correcting accumulated drift and transform misalignment.
- Built a Lanelet2 HD-mapping pipeline from GNSS trajectory data, generating smoothed lane centerlines alongside a separately captured, voxel-downsampled LiDAR reference point cloud.

Multi-Sensor 3D Object Localization (UDrive) | ROS 2, YOLOv8, Python, OpenCV | github.com/nithinkumaraitha/multisensor-3d-localization

- Integrated camera, LiDAR, IMU, and GNSS into a time-synchronized ROS 2 perception stack running at 10 Hz.
- Implemented 2D-to-3D projection recovering object-level (x, y, z) positions from YOLOv8 detections, then transformed local estimates into global GPS coordinates within 0.5 m of GNSS ground truth.
- Quantified drift and cross-sensor alignment error across 12 field runs and documented calibration procedures later reused by the team.

Indoor Autonomous Delivery Robot | ROS 2 Humble, slam_toolbox, Nav2

- Built an autonomous mobile robot completing point-to-point deliveries through multi-room corridors at a 90% success rate over 30 trials.
- Tuned Nav2 costmaps, global and local planners, and recovery behaviors alongside slam_toolbox localization to sustain pose tracking and avoid dynamic obstacles.

DeepLeaf: Plant Disease Detection | Python, TensorFlow, Keras

- Benchmarked MobileNetV2 against ResNet50 on a 54,000-image, 38-class dataset, reaching 96% validation accuracy with 40% less training time.
- Raised held-out accuracy from 91% to 96% under varied lighting through targeted data augmentation and hyperparameter tuning.

Trajectory Planning for KUKA LBR iiwa 7 R800 | MATLAB

- Implemented an inverse kinematics solver and quintic joint-space trajectory generator for a 7-DOF manipulator, verifying continuous position, velocity, and acceleration profiles within joint limits.

ENGINEERING EXPERIENCE

Bharat Heavy Electricals Limited (BHEL), Student Intern | May 2023 - Jun 2023

- Analyzed compressor and turbine assembly workflows across large-scale production lines, mapping process steps, tooling, and handoffs.

Defence Research and Development Organisation (DRDO), Mechanical Engineering Intern | Jun 2021 - Jul 2021

- Executed finite element analysis in ANSYS on 3 mechanical subsystem designs to validate structural integrity under operational load cases.

EDUCATION

University of Delaware, M.S. Robotics | Newark, DE | Expected Dec 2026

Coursework: Robot Dynamics, Autonomous Systems, Computer Vision, Machine Learning, Control Systems

Osmania University, B.E. Mechanical Engineering | Hyderabad, India | Jun 2024